APM466 Assignment 2: Corporate Credit Risk Term Structure

The Toronto-Dominion Bank (TD): A Multi-Model Perspective (2026-2033)

Henry Vianna — University of Toronto — March 23, 2026

1. MARKET-IMPLIED RISK (REDUCED-FORM)

The real-time probability of default (PD) priced in by fixed-income traders was extracted by calculating the exact continuous Yield to Maturity (YTM) for TD's CAD Bond (Maturing 06/2027) via `scipy.optimize.fsolve`, benchmarked against the Government of Canada (GoC) 1-Year T-Bill.

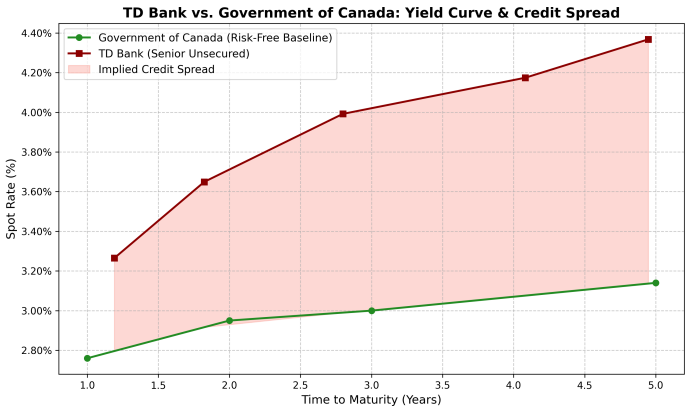

Figure 1: Yield Curve Credit Spread (TD vs. GoC)

Yield & Spread Extraction:

- **TD Bond Dirty Price:** \$102.40
- **Exact Continuous TD YTM:** 3.24%
- **1-Year GoC Benchmark:** 2.76%
- **Implied Credit Spread (s):** 48 bps (0.48%)

Assumptions: A standard senior unsecured recovery rate (R) of 45% is applied.
Output: The 1-Year Market-Implied PD is:

$$PD = \frac{s}{1-R} = \frac{0.0048}{1-0.45} = \mathbf{0.87}\% \tag{1}$$

2. THE MERTON/KMV MODEL (STRUCTURAL)

A structural First Passage Time framework was implemented, treating corporate equity as a European call option on the firm's latent asset value (V).

Calibrated Balance Sheet Parameters:

- **Market Equity (E):** \$214.68 Billion
- **Liability Barrier (D):** \$1.537 Trillion
- **Solved Asset Value (V):** \$1.691 Trillion
- **Solved Asset Volatility (σ_V):** 0.0194
- **Distance to Default (DD):** 7.0200

Assumptions: The risk-free rate is $r = 4\%$. To neutralize unrealistic deterministic drift over multi-year horizons, a dynamic debt barrier growth rate ($\gamma = 4\%$) was introduced.
Output: The theoretical 1-Year PD is strictly bounded near **0.00%**, but scales aggressively over time.

3. TERM STRUCTURE DYNAMICS

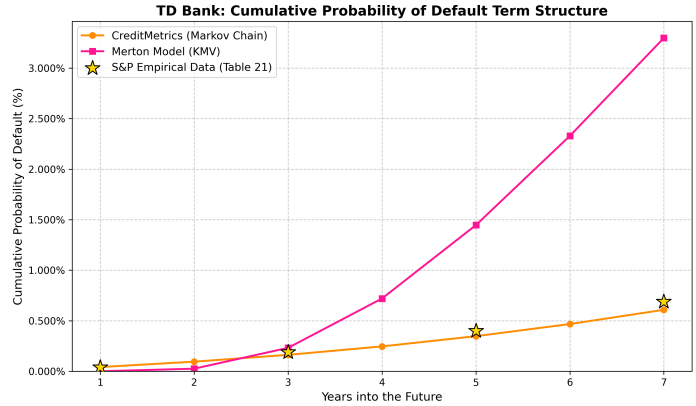

Figure 2: Cumulative Probability of Default (1-7 Years)

The CreditMetrics Baseline (Markov Chain):
By algorithmically normalizing S&P's empirical 1-year transition matrices and applying matrix exponentiation (M^t), a theoretical term structure was generated.

- **Assumption:** Credit migration follows a first-order, memoryless Markov process.
- **Output:** The 1-Year PD for an 'A' rated entity is **0.04%**.

4. STRATEGIC ANALYSIS

The projection highlights a severe divergence between theoretical structural mathematics and empirical survivorship data:

The Bank Leverage Trap: The Merton model projects a highly inflated 7-year PD. This is a mathematical artifact of TD's structural leverage (∼90% debt-to-assets). Because the diffusion process scales uncertainty by $\sqrt{T}$, the pure mathematics assumes the asset "random walk" will eventually breach the debt barrier over long horizons. It artificially inflates long-term risk by ignoring proactive capital management and regulatory intervention.

Ratings Momentum: The divergence between the CreditMetrics projection and S&P empirical observations proves that corporate credit downgrades are not memoryless. Real-world ratings exhibit momentum, resulting in higher multi-year default rates than a pure Markov process predicts.

Conclusion: While structural models effectively isolate balance sheet vulnerability, the Market-Implied spread (∼48 bps / PD: 0.87%) serves as the superior short-term metric. It successfully synthesizes real-time investor sentiment with the "Too Big to Fail" regulatory backstops inherently shielding Canadian SIFIs.